



18/05/2026



Short Course 1

Hands on session:

*From Best-Guess to Better Decisions Modern
Optimisation for Water Distribution Systems*

4th International WDSA/CCWI Conference



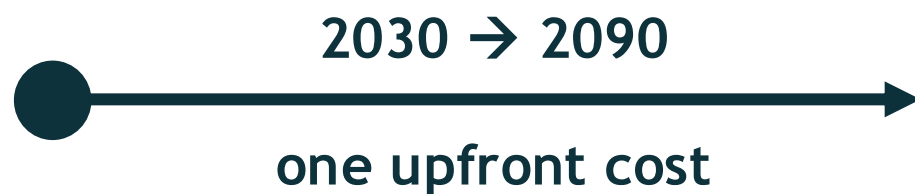
Part 1



Part 2

Traditional Optimisation

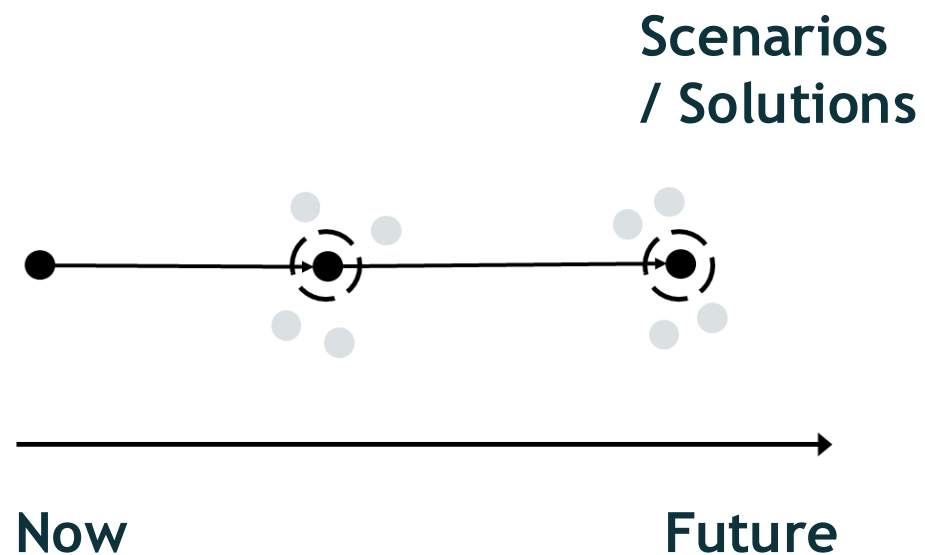
Build all



Goal

Design for best-guess 2090 demand.
Often over- or under-build. Full cost
upfront.

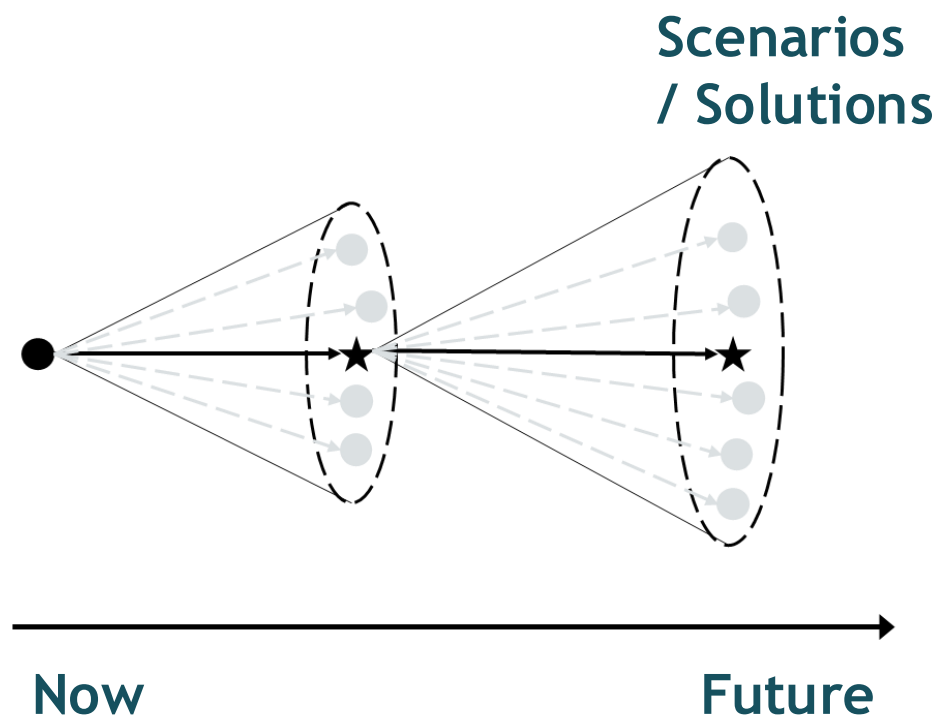
1. Deterministic staged optimization



Goal

Split planning horizon into phases. Find a sequence of interventions with minimum cost. Defer costs into the future. Ignore uncertainty.

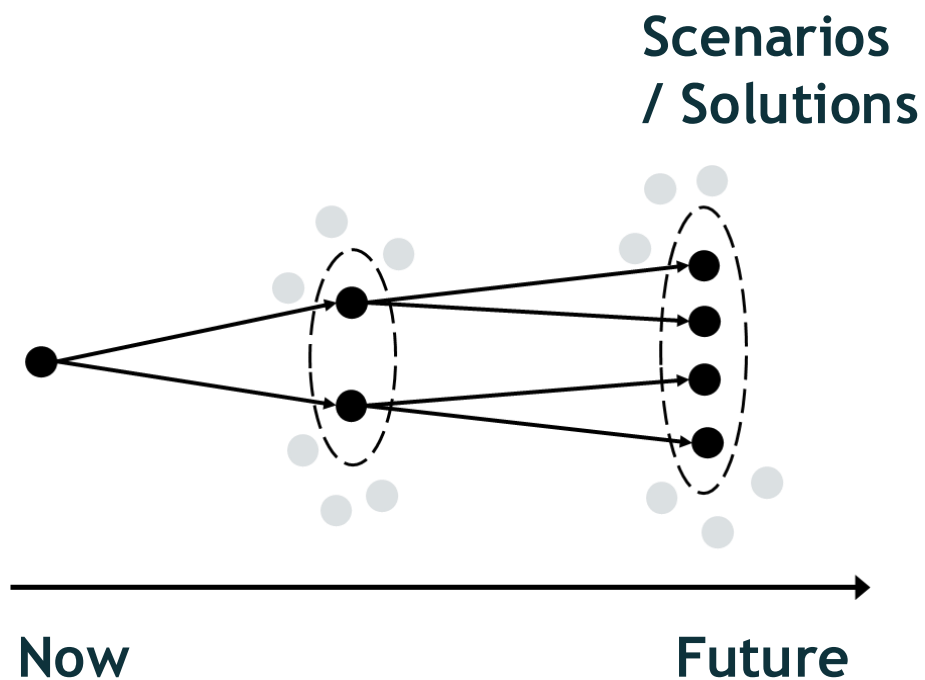
2. Robust staged optimization



Goal

Identify a solution that performs **well enough** under a set of scenarios. The solution is **fixed** for all scenarios. **Pareto front** to balance performance across scenarios. We will **trade cost vs worst-case pressure surplus** in our tutorial.

3. Flexible designs



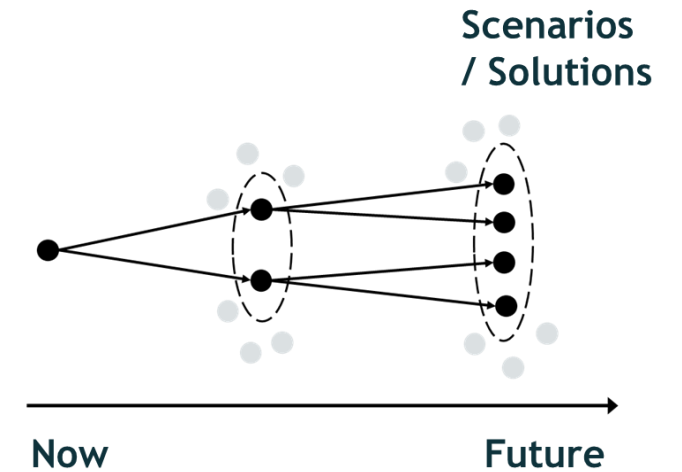
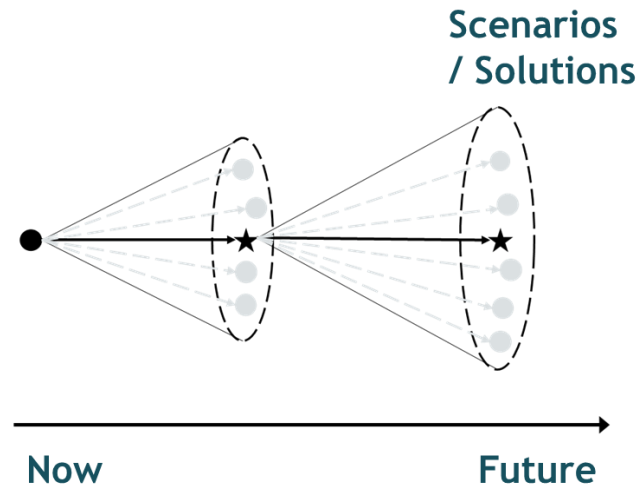
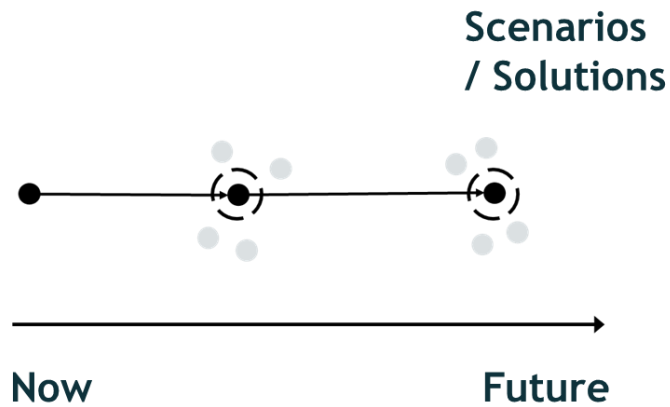
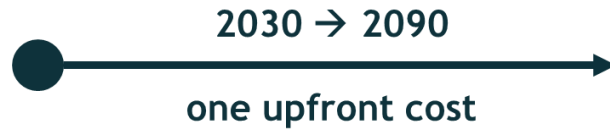
Goal

Adapt when reality reveals itself.
Shared first stage, then branch per
scenario. Minimise expected cost.
Decision tree.



Let's move to the Colab Notebook Part 2

Build all





Part 3